Modelling of BIS-Index Dynamics for Total Intravenous Anesthesia Simulation in Matlab-Simulink ⋆

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Abstract:
The anesthesiologic technique, where substances are injected intravenously, is known as total intravenous anesthesia (TIVA). The anesthesiologist's task is to carefully assess the depth of anesthesia (DoA) and adjusts the injection of intravenous anesthetic agents accordingly. In the paper, we present a framework for studying DoA dynamics within the Matlab-Simulink environment. DoA can be indirectly measured by bispectral index (BIS index), which is calculated from appropriate electroencephalographic (EEG) signals, and is assumed to be influenced primarily by inflow of propofol. The 3-compartmental model is presented and the 4th (virtual) compartment dealing with the effect-site model is introduced. Next, the pharmacodynamic model structure is presented and the BIS-index effect output is formulated. The relevant parameters based on patient's age, weight, height and gender are established. The model is verified by comparing the simulation results to the data obtained during an actual target-controlled anesthetic application of intravenously administered propofol (total duration approx. 70 minutes).
The data were recorded by an Orchestra Base Primea infusion workstation and a Lidco monitor. The data parser and the model developed in the Matlab-Simulink environment provide a basis for further refining the dynamic model and for development and validation of closed-loop control approaches for DoA. The presented model allows conducting simulations and tests of various scenarios of propofol administration within the Matlab-Simulink environment with the goal of a deeper insight into the mechanisms of DoA dynamics. This will lead to better administration methods that will benefit the patient, relieve the workload and allow the anesthesiologist to focus on the critical aspects of the procedure.

Keywords: Target-Controlled Infusion; Propofol; BIS index; Depth of Anesthesia; Matlab-Simulink Environment;

1. INTRODUCTION

When performing a general anesthesia (GA), it is necessary to use substances, which enable deep unconsciousness, analgesia, amnesia and muscle relaxation. A proper introduction of anesthetic agents is essential when performing a diagnostic procedure or a surgery. GA and the related activities in the human body are dynamically very complex processes. The processes involve various pharmacokinetic and pharmacodynamic mechanisms, which have not been fully studied yet.

During the GA, the anesthesiologist needs to monitor the patients vital functions and maintain the functions of vital organs. To achieve adequate GA, substances are introduced in different manners into the patients body. In clinical practice, the most commonly used methods are the intravenous induction of an anesthetic agent, i.e., injection of the anesthetic into a vein, and inhalation induction of anesthesia, whereby the patient inhales the substance from the breathing mixture. The anesthesiologic technique, where substances are injected intravenously, is known as total intravenous anesthesia (TIVA).

The goal of the anesthesiologist is to maintain the appropriate depth of anesthesia (DoA) by adjusting the dosage of anesthetic. Clearly, the pharmacokinetics and pharmacodynamics of the anesthetic agent and the type of procedure must be taken into account. Too deep anesthesia is manifested with a drop in blood pressure level and heart

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rate frequency as well as slow post-operative awakening of the patient from GA. On the other hand, inadequate depth of anesthesia results in the activation of sympathetic nerves, or in the most unlikely event with the patient awakening, which must be avoided at all costs. In modern clinical practice, the DoA is determined by assessing the relevant clinical signs (iris, sweating, movements), by interpreting hemodynamic measurements (see Potočnik et al. (2011)) and by estimating the DoA from EEG signals. The latter is made possible by several established measurement systems, e.g. BIS index, Narcotrend, Scale Entropy and Response Entropy.

Bispectral (BIS) index is a non-invasive measurement method. A BIS monitor is connected to electrodes on the patients head and the bispectral index is calculated from the measured EEG signals. The value represents the DoA. The BIS monitor provides a single dimensionless number, which ranges from 0 (equivalent to EEG silence) to 100. A BIS value between 40 and 60 indicates an appropriate level for GA, whereas a value below 40 is appropriate for long-term sedation due to head injuries. The reference can thus be set to the applicable value; the manner and speed of approaching the reference value depend on the specific characteristics of the procedure and the pharmacokinetics and pharmacodynamics of the substance in the patient's body.

There are various approaches to modelling the effect of propofol described in literature. For these purpose, a number of pharmacokinetic and pharmacodynamic models have been developed, e.g. Marsh et al. (1991), Schnider et al. (1998, 1999), Kataria et al. (1994), Scüttler and Ihmsen (2000), Kenny and White (1990) etc. The models of propofol effect typically define the basic structure of the dynamic system, whereas the parameters depend on the individual patient's characteristics, such as weight, height, age, gender etc., as well as the patient's individual sensitivity to propofol and his ability to excrete propofol.

Certain infusion pumps enable target controlled infusion (TCI), where the pump sets the proper flow of the medication with regard to the model. Various pharmacokinetic models can be employed for this purpose. However, the models often do not reflect the real dynamics, which also depends on individual sensitivity of the patients to the substance, which is typically not considered. Since TCI procedures are based on open-loop induction, they often can not ensure optimal performance, especially when dealing with a particular patient's considerable discrepancy from the mean-population models.

In the paper, we present a framework for studying DoA dynamics within the Matlab-Simulink environment. DoA can be indirectly measured by BIS-index (calculated from appropriate EEG signals). It is assumed DoA is primarily influenced by inflow of propofol, which is set by the infusion pump. The paper is organized as follows. First, we introduce the modelling approach to pharmacokinetics and pharmacodynamics of propofol. The 3-compartmental model is presented. The 4th (virtual) compartment dealing with the effect-site model is introduced. Next, the pharmacodynamic model structure is presented and the BIS-index effect output is formulated. The relevant parameters based on patient's age, weight, height and gender are established. In section 4, the model is verified by comparing the simulation results to the data obtained during an actual anesthetic application of intravenously administered propofol (total duration approx. 70 minutes). In the end, we give some concluding remarks.

2. MODELLING THE PHARMACOKINETICS AND PHARMACODYNAMICS OF PROPOFOL

2.1 The 3-compartmental model

The pharmacokinetics of the derived model is based on the Schnider model (see Schnider et al. (1998) and Schnider et al. (1999)). A similar approach for treating the pharmacokinetics of propofol has been described in Karer (2016). However, in that implementation, the Marsh pharmacokinetic model (see Marsh et al. (1991)) has been used. Furthermore, BIS-index output has not been measured nor treated.

A well-established 3-compartmental model structure, as shown in Figure 1 is used as the basis for dynamic relations.

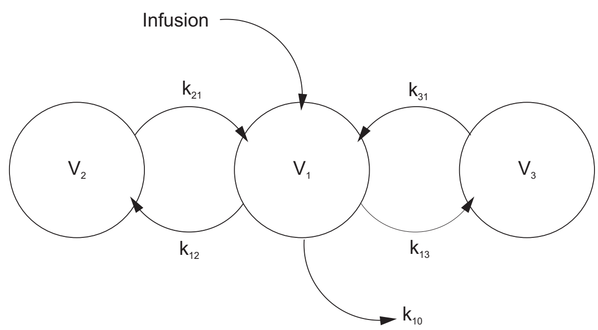

Fig. 1. The pharmacokinetics of the 3-compartmental model.

The 3-compartmental model can be described as follows:

- The drug (namely *propofol*) is injected intravenously into the central compartment (V_1), representing the blood (or plasma) in the body - contained primarily in the arteries and veins and the directly influenced tissues and organs, such as brain, heart, liver, kidney etc.
- The second compartment (V_2) represents the group of tissues that are indirectly affected by the amount of drug in the central compartment, i.e., mainly the muscles. The exchange of the drug with the central compartment is denoted by k_{12} and k_{21}.
- The third compartment (V_3) represents the group of tissues that can store a certain amount of drug, but the exchange with the central compartment is rather slow, i.e., mainly the fat. However, the amount of drug in these tissues influences the amount of the drug in the central compartment in the long run. The exchange of the drug with the central compartment is denoted by k_{13} and k_{31}.
- The drug is eliminated from the body with a rate denoted by k_{10}.

The internal dynamics of the model can be formulated using eqs. (1), (2), and (3).

$$\frac{dx_1}{dt} = \phi - k_{12}x_1 - k_{13}x_1 - k_{10}x_1 + k_{21}x_2 + k_{31}x_3 \quad (1)$$

$$\frac{dx_2}{dt} = -k_{21}x_2 + k_{12}x_1 \quad (2)$$

$$\frac{dx_3}{dt} = -k_{31}x_3 + k_{13}x_1 \quad (3)$$

In eqs. (1), (2), and (3), the variables x_1, x_2, and x_3 represent the concentrations of the drug in compartment V_1, V_2, and V_3. respectively. The infusion flow rate is denoted as ϕ. As noted above, the parameters k_{12}, k_{21}, k_{13}, and k_{31}, represent the partition coefficients that determine the speed at which the drug goes from one particular compartment to another. Finally, k_{10} is the rate of elimination of the drug from the body. Note that the concentration in the central compartment is often referred to as plasmatic concentration $x_1 = c_p$.

2.2 Effect-site concentration model

The effect site for the drug propofol is basically the central nervous system. The effect site is thus part of the central compartment, but the effect of the drug is subject to some dynamics with regard to the (theoretical) concentration in the central compartment. This is mainly due to transportation delay as the drug concentration in the central compartment is not homogeneous, which is evident especially during the transient response, i.e., when the amount of the drug in the central compartment is changing rapidly. The effect-site concentration x_e (often denoted as $x_e = c_e$) is therefore a representation of a volumeless 4th compartment, where the drug is active. This compartment is virtually linked to the central compartment.

Therefore, a 1st order model has been used to describe the effect-site concentration dynamics, as given in eq. (4).

$$\frac{dx_e}{dt} = -k_{e0}x_e + k_{e0}x_1 \quad (4)$$

The virtual link between the central and the effect-site compartment is characterised by the coefficient k_{e0}, which is actually the inverse time constant of the dynamic system describing the connection between the plasmatic concentration and the effect-site concentration x_{e0}.

The schematics explaining the effect-site concentration dynamics are presented in in Figure 2.

2.3 BIS-index effect output

The effect-site concentration of propofol is considered as the main influence on the DoA. Despite acknowledging that DoA is a multivariable and a not very easy to grasp concept, involving deep unconsciousness, analgesia, amnesia and muscle relaxation, we want to keep the model as simple as possible, yet not too simple for our requirements. Therefore, we first assume that BIS index is an adequate measure for DoA. Furthermore, we consider a SISO model, where the input represents propofol inflow, and the output represents the value of BIS index.

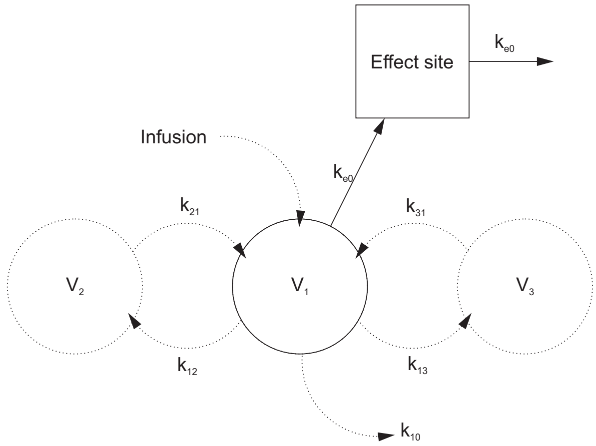

Fig. 2. The effect-site concentration dynamics.

In spite of the fact that pharmacodynamic effect mechanism is not fully studied, usually a sigmoid E_{max} model based on the general Hill equation (see Goutelle et al. (2008)) is considered in the literature, as shown in eq. (5).

$$\begin{aligned} y_{BIS} &= y_{BIS}(x_e) \\ y_{BIS} &= BIS_0 - (BIS_0 - BIS_{min})\frac{x_e^\gamma}{x_{e50}^\gamma + x_e^\gamma} \end{aligned} \quad (5)$$

Here, BIS_0 stands for the characteristic value for a fully awake patient, BIS_{min} is the minimum value of BIS index, and γ is the parameter that defines the nonlinear shape of the response curve. Hence, the combined model can structurally be regarded as a Wiener-type nonlinear model.

2.4 Model parameters

The parameters of the pharmacokinetic model are taken from Schnider et al. (1998), Schnider et al. (1999) and Fresenius-Kabi (2012). The values are given in table 1.

Table 1. Parameter values

Parameter	Value
V_c	4.27 l
k_{10}	0.443 + 0.0107·($weight$/kg - 77) - - 0.0159·(LBM/kg - 59) + + 0.0062·($height$/cm - 177) / min
k_{12}	0.302 - 0.0056·(age/years - 53) / min
k_{13}	0.196 /min
k_{21}	[1.29 - 0.024·(age/years - 53)] * ·[18.9 - 0.391·(age - 53)]$^{-1}$ / min
k_{31}	0.0035 / min
k_{e0}	0.456 / min

Note that the values of model parameters depend on the individual patient's age, weight, height and gender. Parameter LBM is calculated as given in eq. (6).

$$LBM = \begin{cases} 1.1 \cdot weight/\text{kg} - 128(\frac{weight/\text{kg}}{height/\text{cm}})^2 & ;\text{ male} \\ 1.07 \cdot weight/\text{kg} - 148(\frac{weight/\text{kg}}{height/\text{cm}})^2 & ;\text{ female} \end{cases} \quad (6)$$

The parameters of the BIS-index-effect output submodel (see eq. (5)) are set as suggested in Martín-Mateos et al. (2013) and given in table 2.

Table 2. Parameter values

Parameter	Value
BIS_0	95.6
BIS_{min}	8.9
x_{e50}	2.23
γ	1.58

When developing the model in Matlab-Simulink we have to consider the units and the dilution factors of the drug. For example, the parameters expressed in min^{-1} have to be converted to s^{-1}. Furthermore, the input flow of propofol is typically expressed in ml/h, and the dilution is $20mg/ml$, i.e., 2%. The output concentration (plasmatic and effect-site) are expressed in $\mu g/ml$.

3. RESULTS

3.1 Model verification

The Schneider model is also used in the infusion workstation *Orchestra Base Primea* (produced by *Fresenius Kabi* – see Fresenius-Kabi (2012)). We obtained a textual output data file of an actual anesthetic application of target-controlled infusion of intravenously administered propofol, which was recorded by the Orchestra Base Primea infusion workstation during a medical procedure that lasted about 70 minutes. In addition, we also recorded the BIS index trajectory using the Lidco monitor. The data recorded in the output files are as follows:

- The infusion flow of propofol.
- The predicted plasmatic concentration of propofol c_p.
- The predicted effect-site concentration of propofol c_e.
- Value of BIS index.
- Important events (alarms, occlusions, syringe changes, target-value changes).
- Patient data:
  - age (60 years),
  - weight (75 kg),
  - height (168 cm), and
  - gender (female).

A Matlab-based parser was developed. The parser is used to processes the output data files in textual format that are recorded by the Orchestra Base Primea infusion workstation and the Lidco monitor. In this manner we can reformat the data in Matlab-Simulink-environment friendly time-stamped data arrays. Note that the Orchestra Base Primea infusion workstation and the Lidco monitor provide *.txt and *.csv output files, respectively. When parsing the data, it is important to ensure proper data synchronization: that can be done either by synchronizing the internal clocks of the Orchestra Base Primea infusion workstation and the Lidco monitor, or by taking into account the relative delay when parsing the data. In order to verify the model, the parsed data is finally compared to the simulated signals.

3.2 Simulation results

Propofol inflow We used the presented model for simulating the system behaviour with regard to the response to the inflow of propofol influencing the depth of anesthesia through plasmatic concentration and effect-site concentration of propofol. The simulated inflow of propofol was set according to the data recorded by the Orchestra Base Primea infusion workstation. The inflow-signal of propofol $\phi_{propofol}$ was parsed from the recorded data and is shown in Figure 3.

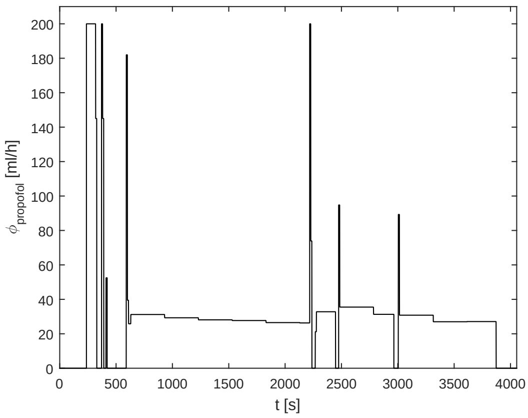

Fig. 3. The inflow of propofol $\phi_{propofol}$.

The presented Matlab-Simulink model was fed the propofol-inflow signal $\phi_{propofol}$ and the resulting trajectories of propofol concentration in the central compartment and in the effect-site compartment (c_p and c_e, respectively) were compared to the data recorded by the Orchestra Base Primea infusion workstation.

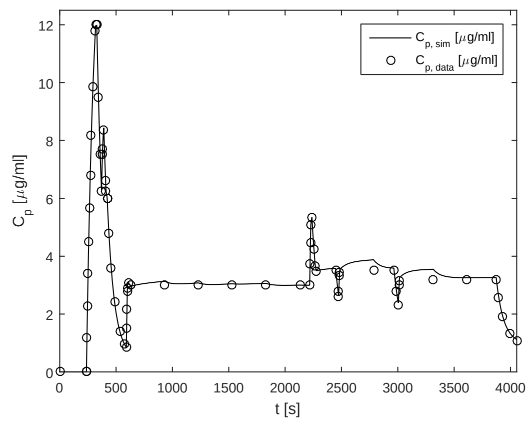

Fig. 4. The simulated plasmatic concentration of propofol $c_{p,sim}$ compared to the parsed data $c_{p,data}$.

Plasmatic concentration The simulated plasmatic concentration of propofol $c_{p,sim}$ trajectory is shown in Figure 4. The simulated results are compared to the parsed data from the Orchestra Base Primea infusion workstation data file $c_{p,data}$.

Effect-site concentration Similarly, the simulated effect-site concentration of propofol $c_{e,sim}$ trajectory is shown in Figure 5. The simulated results are compared to the parsed data from the Orchestra Base Primea infusion workstation data file $c_{e,data}$.

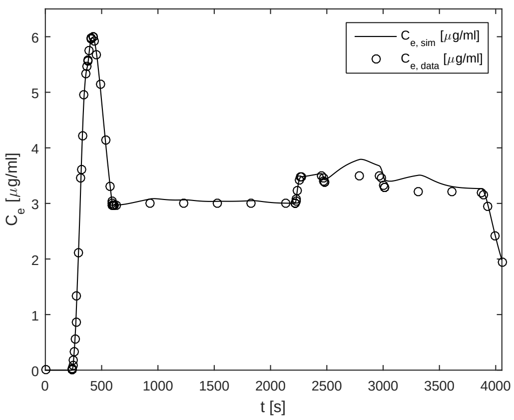

Fig. 5. The simulated effect-site concentration of propofol $c_{e,sim}$ compared to the parsed data $c_{e,data}$.

BIS-index effect Finally, DoA is assessed using BIS index. The simulated BIS-index trajectory BIS_{sim} is shown Figure 6. The simulated results are compared to the parsed data from the Lidco monitor data file BIS_{data}.

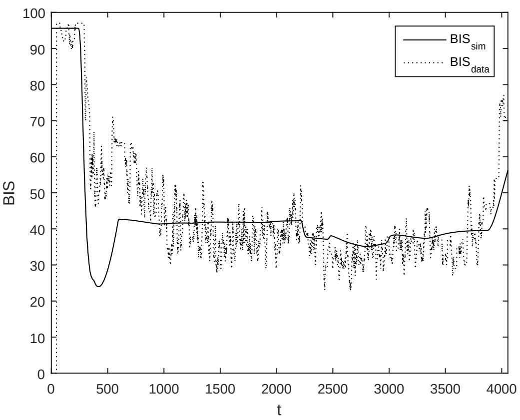

Fig. 6. The simulated BIS-index trajectory BIS_{sim} compared to the parsed data BIS_{data}.

3.3 Model quality evaluation

The simulated results are compared to the parsed data so as to evaluate the quality of the presented model. Some of the well-established quantitative measures for predictive quality are *prediction mean square error* (PMSE), *median performance error* (MDPE), and *median absolute performance error* (MDAPE) (see Mertens et al. (2003)), which are calculated according to eqs. (7), (8), and (9), respectively.

$$PMSE_x = \frac{1}{N}\sum_{i=1}^{N}(x_{i,sim} - x_{i,data})^2 \tag{7}$$

$$MDPE_x = median\{\frac{x_{i,data} - x_{i,sim}}{x_{i,sim}} \cdot 100\%\}_{i=1,\dots,N} \tag{8}$$

$$MDAPE_x = median\{|\frac{x_{i,data} - x_{i,sim}}{x_{i,sim}}| \cdot 100\%\}_{i=1,\dots,N} \tag{9}$$

Here, x_{sim} and x_{data} stand for the simulated and the "measured" data, respectively, and N is the number of data points in the data set.

For our particular case, the values of the aforementioned criteria are presented in table 3.

Table 3. Predictive quality measures

Signal x	$PMSE_x$	$MDPE_x$	$MDAPE_x$
c_p	$19.6 \cdot 10^{-3}$	-1.69%	1.73%
c_e	$14.8 \cdot 10^{-3}$	-1.73%	1.74%
BIS	223	-3.35%	12.3%

The criterion functions concerning the pharmacokinetic part of the model (with regard to signals c_p and c_e) suggest low discrepancy between the simulated and the parsed data, which was expected since the same model is used in both cases. The differences probably occur due to numerical issues (e.g. integration method, solver) and inaccurate recording of the propofol inflow $\phi_{propofol}$ in the Orchestra Base Primea infusion workstation.

On the other hand, the pharmacodynamic submodel dealing with the response of the BIS-index signal to the propofol effect-site concentration c_e is a more complex issue. Again, the simulated data was compared to the parsed data. The simulation error arises for various reasons.

- First, the propofol effect-site concentration c_e can not be measured directly. The simulation results are based on the simulations of the pharmacokinetic model, which is not necessarily accurate.
- Besides the inflow of propofol, the BIS-index value depends partly on the inflow of remifentanil (analgesic), which is also administered intravenously during the procedure. However, the SISO model does not take this influence into account.
- What is more, various events during the procedure, e.g. incision, can impact the value of the BIS index, which is not considered in the model.
- In addition, the BIS-index signal is subject to considerable noise. Therefore, it is even more difficult to match it in simulation, which is the reason for the relatively high figures, especially the MDAPE value.
- The BIS-index value is calculated online from a window of past samples. Therefore, the signal on the monitor is inherently delayed with regard to the actual DoA.
- The parameters shown in table 2 are suggested for the maintenance phase of anesthesia, whereas in this case, the whole anesthetic procedure is considered. If only the maintenance phase is taken into account, the predictive quality measures improve considerably, as given in table 4 (row $BIS_{maintenance}$).
- Furthermore, the parameters are derived based on a large population and do not consider the specific

sensitivity of the particular patient to propofol. The parameters x_{e50} and γ are adjusted to the treated patient, which can be done a posteriori by optimization. In our case, the optimization was conducted using the Nelder-Mead Simplex Method. The resulting parameters are $x_{e50} = 1.97$ and $\gamma = 1.50$. In this manner, the results are improved even further, as shown in table 4 (row $BIS_{maintenance,opt}$) and Figure 7.

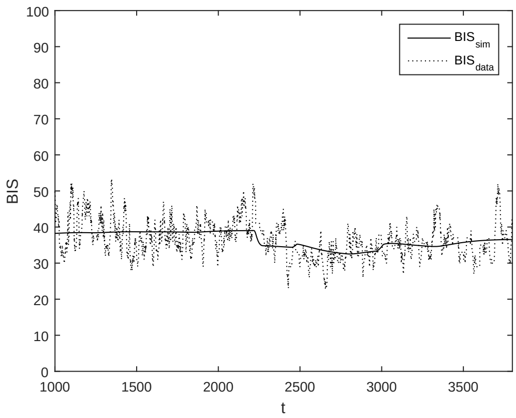

Fig. 7. The simulated BIS-index trajectory BIS_{sim} compared to the parsed data BIS_{data} - maintenance phase.

Table 4. Predictive quality measures

Signal x	$PMSE_x$	$MDPE_x$	$MDAPE_x$
$BIS_{maintenance}$	29.7	−8.52%	10.6%
$BIS_{maintenance,opt}$	20.9	−1.11%	8.29%

4. SUMMARY

The data parser and the model developed in the Matlab-Simulink environment represents a useful starting point for further researching the DoA dynamics. The pharmacokinetic part of the model, namely the c_p and c_e submodels, has been taken from the literature and implemented in the Matlab-Simulink environment. The drawback is that it is not possible to properly validate the c_p model without actually measuring the plasmatic concentrations of propofol, which can not be done continuously, but only by taking and analysing consecutive blood samples. What is more, the effect-site concentration c_e can not be physically measured at all.

The validation of the pharmacodynamic submodel is even more demanding. It is based on a non-measurable input, exhibits a multivariable nature, is susceptible to various events during the procedure, its output-signal measurements are inherently delayed w.r.t. DoA, the dynamics change during different phases of the procedure etc. In spite of this, the model can be used to simulate the actual DoA. Furthermore, we have shown that the pharmacodynamical model can be improved by replacing the population-based parameters with individually optimized ones. That said, the adequacy of the structural properties of the model, which rely on the sigmoid E_{max} model based on the general Hill equation, has not been considered and remains one of the tasks for future research.

To sum up, the presented framework provides a basis for obtaining a suitable model for development and validation of closed-loop control of DoA. The presented model allows simulations and tests of various scenarios of propofol administration within the Matlab-Simulink environment with the goal of a deeper insight into the mechanisms of DoA dynamics, which will lead to better administration methods that will benefit the patient and relieve the anesthesiologist's workload. The frameworks represents the first steps of our joint future research in collaboration with the Department of Anaesthesiology and Surgical Intensive Therapy, University Medical Centre Ljubljana.

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